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If $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$, then calculate the inverse of A^t (if exists).

Answer ([Please click here to Add Answer](#))



Evaluate the determinant $\begin{vmatrix} 2 & 4 & 8 & 16 \\ 0 & 3 & 6 & 12 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & -3 \end{vmatrix}$.

Answer ([Please click here to Add Answer](#))



If $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$, then calculate the inverse of AB (if exists) ; where $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 4 & -5 \\ -3 & 4 \end{bmatrix}$.

Answer ([Please click here to Add Answer](#))





Let S be a parallelogram determined by the vectors $\vec{b}_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\vec{b}_2 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $A_1 = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$. Compute the area of the image of S under the mapping $\vec{x} \rightarrow A_1 \vec{x}$.

Answer ([Please click here to Add Answer](#))



Find an LU - decomposition of the matrix $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$.

Answer ([Please click here to Add Answer](#))



If $A = \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & -3 \\ 9 & -3 & 3 \end{bmatrix}$, then find the minors of first row and determinant of A using these minors .

Answer ([Please click here to Add Answer](#))



Answer (Please [click here](#) to Add Answer)

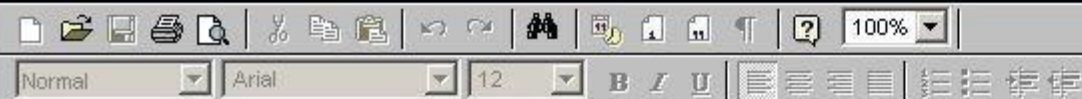
If $A = \begin{bmatrix} 4 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}$, then determine whether $\vec{u} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ is in $\text{Nul } A$ or not ?

Answer ([Please click here to Add Answer](#))



Let W be the set of all vectors of the form $\begin{bmatrix} 3a \\ 2a \\ a \end{bmatrix}$. Find a vector $\vec{v}$ in $\mathbb{R}^3$ such that $W = \text{Span}(\vec{v})$.

Answer ([Please click here to Add Answer](#))



$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 6 & 3 & 4 \end{bmatrix}$$

Show that the coefficient matrix of the following linear system is strictly diagonally dominant. Also calculate the second iteration using Gauss – Seidel method if the first iteration is

$$(x_1^{(1)}, x_2^{(1)}, x_3^{(1)}) = (1.3, 1.04, 0.936)$$

$$10x_1 + 2x_2 + x_3 = 13$$

$$2x_1 + 10x_2 + x_3 = 13$$

$$2x_1 + x_2 + 10x_3 = 13$$

Answer ([Please click here to Add Answer](#))



Show that $\begin{vmatrix} a+h & a-h & d \\ b+k & b-k & e \\ c+l & c-l & f \end{vmatrix} = 2 \begin{vmatrix} a & a-h & d \\ b & b-k & e \\ c & c-l & f \end{vmatrix}.$

Answer ([Please click here to Add Answer](#))



If a matrix A is decomposed into $L = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & -7 \\ 0 & 1 \end{bmatrix}$ by LU factorization, then find A .

Answer ([Please click here to Add Answer](#))



If $A = \begin{bmatrix} 4 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}$, then determine whether $\vec{u} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ is in $\text{Nul } A$ or not ?

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Construct partitions of the following matrix into three 2×2 blocks :

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 6 & 3 & 4 \end{bmatrix}$$

Answer ([Please click here to Add Answer](#))



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Answer ([Please click here to Add Answer](#))



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Answer ([Please click here to Add Answer](#))



Evaluate the determinant

$$\begin{vmatrix} 2 & 4 & 8 & 16 \\ 0 & 3 & 6 & 12 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & -3 \end{vmatrix}.$$

Answer ([Please click here to Add Answer](#))



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